Kema Sai Vedhamsh

ee24btech11211@iith.ac.in | ksvedhamsh@gmail.com | +91 8247235214 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.68
FIITJEE	Class XII (SSC)	2024	94.70%
FIITJEE	Class X (SSC)	2022	10.00%

ABOUT ME

I am an undergraduate at IIT Hyderabad with a fascination for semiconductor device physics and quantum transport. I strive to build a first-principles understanding of physical systems, aiming to approach research with patience and analytical discipline. Whether grappling with complex transport mathematics or developing numerical solvers, I believe in learning through persistent effort and rebuilding from the ground up when necessary. My goal is to continuously refine my research capabilities as I work toward pursuing a PhD in carrier transport dynamics.

PROJECTS

Research Work - Exploring Multi-barrier Superlattice structures (**RTD**) for **PUF** applications - *Github*

- Using the 1D NEGF solver, simulated the transfer characteristics of Multi-barrier devices
- The variablity of these devices is measured in terms of the peak/valley currents and NDR
- Obtained CDF plots with respect to the peak resonance currents
- Characterized devices to bits, by comparing their peak resonance currents with a reference current
- Developed a PUF that utilizes this data and has a large CRP set
- Analyzed this PUF using metrics like Diffuseness and Uniqueness
- Analyzed what parameters give ideal values of these metrics

Quantum Transport Solver - Implementing **NEGF** formalism to simulate **1D transport** in quantum devices - *Github*

- Developed a custom Python-based NEGF transport solver
- The solver evaluates device parameters under two modes of transport
- The Ballistic mode -
- Developed using the standard Tsu-Esaki model
- Calculates the needed Greens function matrix columns using a custom, optimized version of Thomas algorithm
- Calculates the Contact self energies and Transmission coefficients at certain biases
- Calculates the current density using Transmission coefficients after sweeping the device across biases
- The Dissipative mode -
- Implemented Dimitri Nikonovs model to include the transverse moments in the unconfined directions for scattering effects
- Implemented optical and acoustic phonon scattering
- Calculates the required terms using RGF for Retarded and Lesser/Greater Greens matrices
- Implemented Phonon self energy calculation according to Nikonovs model
- Calculates the Electron/hole densities using the Lesser/Greater Greens matrix
- Implementing Current calculation using Meir-Wingreen formalism
- Implemented Self consistency by using Poisson's equation using Newton-Raphson method for convergence

Device Calibration - Used In-house tool to calibrate a **p-fet Nanowire** to match research papers - *Github*

- Constructed a pfet Nanowire according to the specifications given in research work
- Calibrated the parameters of this device to match the transfer characteristics obtained in the paper
- Demonstrated a new central gate pfet nanowire which gives better transfer characteristics than classic pfet
- The central gate pfet increased the on-state current by approximately twice, due to the extra channel
- The central gate pfet also gave better off-state characteristics, sub-threshold swing approaching the Ideal at 300K

SKILLS

Languages - C, Python, Verilog

EDA & Simulation - LTspice, NGspice, EDR CAD, Solid Edge

PCB Design - KiCad, Altium

RELEVANT COURSES

Core Theory - Semiconductor Device Fundamentals, Numerical Methods for Device Modelling, Modelling of Nanoscale MOSFETS,

CMOS Processing, Solid State Devices, Introduction to VLSI Design

Core Lab - Semiconductor Devices Lab, Electrical Circuits lab, Analog devices lab

POSITIONS OF RESPONSIBILITY

TA - for courses "Numerical Methods for Device Modelling" and "Engineering Electromagnetics"

- Conducting Doubt Solving sessions
- Teaching concepts of NEGF
- Advising over student projects